

Analytical architecture of the study

From raw source files to final validation

DATA ASSEMBLY · arithmetic, not inference

Stage 0 Source ingestion

FC-16 Vol. II Ch. 5 annexures · DPIIT state-wise FDI · MoSPI GSDP (2011-12 base) · Census 2011 Primary Census Abstract

§ 4.2 – 4.6

Stage 1 Panel construction

Balanced 10×5 panel · ratios as per cent of GSDP · $\ln(\text{FDI})$ · year-on-year GSDP growth · COVID dummy · 48 usable FDI cells of 50

§ 4.1, 4.3 – 4.5

Stage 2 Composite discipline ranking

Four fiscal ratios ranked separately across the ten states, then averaged into a composite score, year-wise and over five years

§ 4.7, 6.2

BRANCH A · classical

Stage 3 Panel estimation

Models 1 and 2, each estimated twice: fixed effects with Driscoll-Kraay errors (within-state), and pooled OLS with Census controls

§ 5.1, 6.3 – 6.4

Stage 4 Diagnostics

Variance inflation factors · pairwise correlation matrix · poolability F-tests

§ 6.5

Stage 5 Subsample perturbation

Refits dropping the COVID year, Uttar Pradesh, and Jharkhand in turn

§ 6.5

BRANCH B · machine learning

Stage 6 Regularised fits

Five time-varying regressors only. LASSO at a leave-one-out cross-validated penalty · Elastic Net · random forest read via SHAP

§ 5.2, 6.6

Stage 7 Leakage audit

Urbanisation diagnosed as a disguised state identifier; branch refitted without time-invariant controls

§ 5.2, 6.6

Stage 8 Stability layer

Ten leave-one-state-out LASSO refits · LOO-CV out-of-sample R^2 for all three

§ 6.6

CONVERGENCE

Stage 9 Convergence test — FINAL VALIDATION

Five methods sharing nothing beyond the same 48 observations: pooled panel regression · Spearman rank correlation · LASSO · Elastic Net · SHAP importance. A result is accepted only where the sign and the rank agree across both branches, and disagreements are reported.

§ 6.6, 7.1

Stage 10 Residual case analysis

Jharkhand and Uttar Pradesh examined by name as departures from the fitted pattern: evidence about where the pattern stops holding

§ 7.3